

Ayoconnect: Enabling businesses to deliver API- powered financial products quickly and reliably

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[About Ayoconnect](#)



About Ayoconnect

With Google Cloud, Ayoconnect is realizing its ambition to provide digitized financial products in a standardized way to businesses across Southeast Asia.

Founded in 2016 and headquartered in Singapore, Southeast Asia, Ayoconnect is a SaaS company that builds an open finance platform for developers to select from a range of financial products and deliver them to their customers. By making direct debit, bill payment, subscription management, and data verification services available via APIs, Ayoconnect aims to build an infrastructure that enables clients to monetize and expand their businesses quickly and efficiently through open finance.

Google Cloud results

- Enables the business to keep infrastructure team headcount at three versus 10 to 12
- Achieves uptime of 99.8% to support near always-available requirement for financial services SLAs
- Ensures automated scalability in minutes to support campaigns and events that stimulate traffic
- Empowers the business to support

Support 1000+ in AP Apigee

Industries: Financial Services & Insurance

Location: Indonesia

(<https://cloud.google.com/monitoring/>), [Cloud](#)

searche

About Searche

Searche is a cloud consulting business that specializes in cloud data engineering, AI and machine learning, as well as advanced cloud infrastructure technology such as Anthos and Kubernetes. The business has moved 3,000+ clients successfully to the cloud, and is part of the Google Cloud partner network.

Google Cloud (<https://cloud.google.com/>)

Apigee API Platform
(<https://cloud.google.com/apigee/>)

Monitoring (<https://cloud.google.com/monitoring/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Searche is a cloud consulting business that
Cloud Storage (<https://cloud.google.com/storage/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

the
Indonesian
Government
and Bank of
Indonesia's
push to
standardize
and digitize
financial
services
through open
finance APIs

Cloud IAM (<https://cloud.google.com/iam/>) has
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Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Virtual Private Cloud
(<https://cloud.google.com/vpc/>)

VPC Service Controls
(<https://cloud.google.com/vpc-service-controls>)

Founded in 2016, [Ayoconnect](https://www.ayoconnect.id/?lang=en)
(<https://www.ayoconnect.id/?lang=en>) helps financial
institutions, startups and businesses launch
financial products quickly, with its open finance
platform, making direct debit, bill payment,
subscription management and data verification
services available via APIs. The business employs
270 people spread across six countries.

"We emerged from an explosion of innovation in
financial services in Indonesia, and a recent focus by
the Indonesian Government and Bank of Indonesia
on digitization and standardization across the
sector," explains Adi Vora, Co-Founder and CPTO,
Ayoconnect. "We work with banks, regulators and
clients to identify gaps in the market, and find ways
to close them with products that help deliver the
financial services vision for the country as a whole."

Ayoconnect adopted its current business-to-
business model after its success in the business-to-

consumer space. "We were pivoting from a business-to-consumer application called Ayopop that enabled people to pay bills and complete other transactions, to a business-to-business model that solved a range of payments use cases," says Vora. "Through standardization and visibility, we could solve the issues created by multiple infrastructure types and API formats across financial services."

Ayoconnect was provided with limited options when it first decided to launch the open finance platform on a cloud environment, as regulations required Indonesians' financial and transactional data to stay within the country. For that reason, the business started developing and running products on a domestic cloud service. However, this provider could not fully meet Ayoconnect's requirements, forcing the business to start building its products from scratch. "We did good work launching a product in a distributed Kubernetes environment on that service, but we experienced issues with performance and stability," explains Vora.

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*—Adi Vora, Co-Founder
and CPTO, Ayoconnect*

**Ensuring data sovereignty
compliance in Indonesia**

With Google Cloud (<https://cloud.google.com/>)'s launch in Jakarta, Indonesia in 2020, Ayoconnect was able to comply with the country's data sovereignty requirements while leveraging several of Google Cloud products, services, infrastructure, and expertise. The Ayoconnect team identified Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) as providing the automation, scalability, and agility that would meet its needs, while Cloud SQL (<https://cloud.google.com/sql>) would provide the required database stability and performance.

"We held a number of calls with the Google Cloud team in Singapore to understand the features, functionality and capabilities of Google Cloud products and services, and we received some migration credits that enabled us to test performance and scaling before we migrated to production traffic," says Vora. "This convinced us that Google Cloud was the answer to our requirements."

Ayoconnect then adopted a Google Cloud architecture centered on GKE and Cloud SQL, with Bitbucket pipelines used to automate the deployment of code and security scripts. In addition, Identity and Access Management

(<https://cloud.google.com/iam>) enabled Ayoconnect to provide its teams with role-based access to data and systems, helping ensure its security systems and processes met the requirements of enterprises operating in highly regulated industries such as financial services.



Ayoconnect Isometric

Ayoconnect turns to Searce to deploy Apigee

When Ayoconnect decided to augment its business-to-business portfolio with direct debit and subscription management services, it turned to technology consulting firm Searce

(<https://www.searce.com/>) and the Apigee API management platform. "We had attended an event run by Searce just before the pandemic, and the opportunity to work with its team emerged when we decided to launch our direct debit solution," explains Vora. "We conducted due diligence, including checking their technical and non-technical credentials, and we were impressed by their work building an open banking system for one of our partners, which aligned with our own roadmap."

With the Searce team demonstrating their in-depth knowledge of the Indonesian landscape, and expertise in Apigee API Management

(<https://cloud.google.com/apigee>), taking the next step and engaging them to help develop Ayoconnect's

new products with the API platform integrated was an easy one. Searce helped Ayoconnect determine the architecture that would best enable its use cases, and continues to provide ongoing support as the business aims to ensure its solutions can meet its needs for at least the next three to five years.

Apigee now authenticates Ayoconnect clients to access the business's API products. "Apigee has helped us set up proxies, maintain a developer portal, and monitor our APIs," explains Vora. "It also provides a very good security layer with the architecture we have set up with Searce."

"Apigee connects to developing, staging and production, which is hosted in a separate Virtual Private Cloud (<https://cloud.google.com/vpc>) linked by VPC bridges. This ensures no clients directly hit the environments in which services are hosted."

The architecture also incorporates Cloud Run (<https://cloud.google.com/run>)'s fully-managed serverless platform for delivering containerized applications used to run Ayoconnect's Apigee-powered open banking system. The business is migrating a number of queues to an event-driven architecture based on Pub/Sub (<https://cloud.google.com/pubsub>). Cloud Monitoring (<https://cloud.google.com/monitoring>) enables the business to monitor its infrastructure and applications to ensure potential incidents can be addressed proactively rather than reactively, minimizing risks. Meanwhile, Cloud Storage (<https://cloud.google.com/storage>) enables the business to store and archive data. With Searce, the

business is also looking at the opportunities that Cloud Spanner (<https://cloud.google.com/spanner>)'s distributed SQL database management and storage service present to the business.

"Apigee has helped us set up proxies, maintain a developer portal, and monitor who is hitting which API. It also provides a very good security layer with the architecture we have set up with Searce. Apigee connects to developing, staging and production, which is hosted in a separate Virtual Private

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*—Adi Vora, Co-Founder
and CPTO, Ayoconnect*

Moving from manual requests to seamless scalability

Moving away from its incumbent cloud provider in Indonesia to Google Cloud enabled Ayoconnect to accelerate and automate processes to scale its cloud infrastructure in line with demand. Previously, if Ayoconnect required a change, it had to send a manual request that could take up to six hours to process, meaning the business could not respond quickly to changing circumstances. For example, when a client's promotion generated transactions and user numbers more than twice those forecasted, Ayoconnect could not scale up quickly, disrupting the user experience until four hours into the campaign when the business was able to add resources to support demand.

Now, the business can seamlessly scale to support client promotional campaigns and other activities that generate high traffic. "With Google Cloud, our system takes a few seconds to increase or decrease as required, and takes care of itself," explains Vora.

Successfully executing the transition from B2C to B2B

With Google Cloud, Ayoconnect has successfully executed its transition from a business-to-consumer to a fast-growing business-to-business provider. The business has grown from two clients in 2019 to 130 to date, including several big technology companies and banks. API hits are now 1000 times over the number of hits Ayoconnect experienced as a business-to-consumer organization.

The business is achieving this growth while delivering the stability and availability required by enterprise-grade customers in highly regulated industries. Uptime in particular is critical to Ayoconnect, its clients and their end-users. Vora notes that financial APIs "cannot ever go down" as customers pay bills and transact 24 hours a day, seven days a week. "Uptime is a key service level and with Google Cloud we are close to 99.8%, well within our business requirements," says Vora.

"With API management and a scalable, reliable infrastructure from Google Cloud, we have executed our plan to help deliver standardized and digitized financial services to businesses and their end-users in Indonesia through open finance."

*—Adi Vora, Co-Founder
and CPTO, Ayoconnect*

Keeping team headcount in single figures

As a fast-growing but still early-stage business, Ayoconnect needs to control costs and ensure its comparatively scarce resources are allocated efficiently. In its previous cloud environment, the infrastructure team had to add new nodes and virtual machines manually, keeping them from activities that could deliver more value to the business. Now, through architecture optimization enabled by Google Cloud, Ayoconnect can keep the team lean and its maintenance and development costs low.

"Our infrastructure team is just three people and the automation and optimization of Google Cloud enables us to keep headcount at that level," explains Vora. "Without these capabilities, we would need a team of 10 to 12 people to do everything we do now."

With its Google Cloud architecture now well established, Ayoconnect is now focusing on optimizing some legacy systems and working with Google Cloud and Searce to understand how to move them to the cloud, with cost optimization and maximizing the value Apigee can deliver to the business. "With API management and a scalable, reliable infrastructure from Google Cloud, we have executed our plan to help deliver standardized and digitized financial solutions to businesses and their end-users in Indonesia through open finance," concludes Vora.